

CHAPTER 6:

# Rail-Trails and Rails-With-Trails

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## 6.1 Introduction

Rail-trails (RT) and rail-with-trails (RWT) are a unique type of shared use path. Rail-trails use inactive, abandoned or railbanked railroad corridors. Rail-with-trails share a corridor with an active railway. To simplify the terminology, in this chapter the term rail-trail is used to reference both facility types, and rail-with-trail applies only to active shared railroad corridors.

Like shared use paths, rail-trails are used by bicyclists and pedestrians, including walkers, joggers, runners, people with baby strollers, skaters, and people walking dogs. During winter months, rail-trails may be used for cross country skiing, snowshoeing and snowmobiling, where permitted. The facilities should take into account the various capabilities, needs and characteristics of a wide range of users.

This chapter covers the planning and design of rail-trail conversions. Not included are issues relating to insurance, liability, lease agreements with rail-trail managers, or the operation of rail-trails.

### 6.1.1 Characteristics of Rail-Trails

Many of the characteristics of shared use paths (refer to Section 5.2.1) are also common to rail-trails and rails-with-trails. These include: continuous separation from motor vehicle traffic; frequent access points; increased levels of safety and security; scenic qualities; connectivity to a variety of land uses; well-designed street and driveway crossings; potential for shorter trips; activity; uniform design and good engineering; informative signing and marking; context sensitive design and aesthetics; regular inspection and maintenance and potential for making an economic contribution to the community.

In addition, rail-trails have many unique characteristics that set them apart from shared use paths:

- Ownership, operation and maintenance.
- Existing infrastructure.
- Alignment.
- Grade.
- Geometrics.
- Drainage.
- Scenery.
- History.
- Distance markers.
- Community support.
- Remoteness.
- Contamination and deterioration.
- Potential for making an economic contribution to the community.

1. **Ownership, Operation and Management.** The issues surrounding ownership, operation and management of a rail corridor, whether active or inactive, are often complex. If a corridor is intact, it may be owned or leased by a governmental entity or by private interests. Use of the corridor for railroad or trail

Swanton – Missisquoi Valley Rail Trail



Swanton – Missisquoi Valley Rail Trail



Burlington – Burlington Bike Path



**Rail-trails and rail-with-trails are a unique type of shared use path. Like shared use paths, rail-trails are used by bicyclists and pedestrians. During winter months, rail-trails may be used for cross country skiing, snowshoeing and snowmobiling.**

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purposes can be the responsibility of the corridor owner or it may have been legally assigned to others through a lease, easement, cooperative agreement or finance and maintenance agreement.

If ownership of a former railroad corridor has reverted to adjacent landowners, each landowner will have a say as to whether the land may be used for trail purposes. Whatever the case, expect that all interested parties will need to be involved in the development of the trail.

For state owned railroads, historic deed information appears on valuation sheets maintained by the Rail Division at the Agency of Transportation. Be cautious when using valuation maps, however, because they may not have been fully updated to reflect current ownership. Where ownership is in question, consult deeds, town property records or other resources that may have been produced when the corridor was constructed.

2. **Existing Infrastructure.** The sub-base, superstructure, drainage, bridges, and crossings are already established. Design is often a matter of working with or adapting the existing infrastructure to meet the needs of a rail-trail.
3. **Alignment.** The alignment of a rail corridor is already fixed and known. Except for linkages to other parts of the pedestrian and bicycle network, or for certain roadway or water crossings, there is little uncertainty about the rail-trail location. Rail-trails are typically “straighter” than shared use paths to accommodate the needs of trains, a characteristic that can be used to advantage in connecting destinations.
4. **Grade.** Railroad grades are limited to 1:20 (5 percent) or less. Grades of 1:40 (2.5 percent) are more often the norm. As a result, the grades along a rail corridor are usually more moderate than a shared use path. This makes rail-trails appealing to many walkers and bicyclists, and somewhat easier to adapt to ADA guidelines.
5. **Geometrics.** The steep side-slopes and generous widths and clearances of a rail corridor — designed for large rail equipment — can be cost effectively adapted for use as a rail trail. These geometric features should be maintained because there is always the possibility of returning an inactive railroad corridor to active rail service.

*Side slopes and shoulders:* To maintain a moderate gradient, railroad corridors were often constructed with taller fills and deeper cuts. Depending on the material used to support the loads of a railway facility, existing side slopes along rail embankments may be as steep as 2:1.

*Horizontal widths and clearances:* Right-of-way widths vary greatly but average from 4 to 5 rods (20 to 25 m [66 to 83 ft]). Horizontal clearances for train use vary depending on the curvature or lack of curvature of the track. All fixed objects should be kept well outside the dynamic envelope of the train.

*Vertical clearances:* The standard vertical clearance over a railroad track is 7 m (23 ft), measured from the top of the rail. If converted to a trail, clearance for equestrians and snowmobile grooming equipment should also be considered (when these uses are permitted).

6. **Drainage.** Rail beds were built to exacting design specifications to direct and move water away from them. When the corridors are in active use, railroad crews regularly inspect and keep side areas clear, drainage ditches cleaned out, drainage ways open, and water moving. Without an aggressive maintenance program, vegetation may take over, drainage may become blocked, beavers can

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create dams that cause water to backup, or property owners may modify drainage at crossings. Ongoing maintenance programs are required to preserve and protect the corridor from the elements (refer to Chapter 10, Maintenance).

7. **Scenery.** Because the grade of a railroad needs to be as level as possible, railroad corridors were often located along natural features such as waterways and valleys where the grades are most level and the scenic attributes can be exceptional.
8. **History.** Evidence of prior railroad activity can still be found along a railroad corridor or in the collections of railroad enthusiasts. It's important to utilize such artifacts to recall early days of transportation. Relics suitable for a rail-trail project can include: mile markers; whistle signs; antique grade crossing signs; grade crossing signal relay cabinets; signal bridges; semaphores; telegraph poles and insulators; stations and depots; workers' shelters; storage warehouses; stone abutments; and viaducts, bridges and trestles.

In addition to these remnants, rail-trails also traverse parts of cities and the countryside that highlight bygone eras of development and industrialization, such as factories housed in brick buildings, as well as silos and barns in agricultural areas. These structures, and the corridors they occupy, epitomize the progress of American industry, engineering and labor, thus educating trail users about the route's history. When combined with the stories of people who built them, these features give a rail-trail a "sense of place," strengthening a user's understanding of, and connection to, the trail and its surrounding region.

Photographs, maps, advertisements, timetables and brochures can also be used to generate interest and user support in rail-trail projects.

9. **Distance markers.** All railroad corridors have mile markers, whether physically located along the corridor, indicated on land records and valuation sheets, or both. However, unlike distance markers recommended for shared use paths, which usually begin at a trailhead, mileage indications for railroad use may begin miles away from a trailhead or even in a distant state or country. Railroad mile markers have legal as well as historic significance and should not be removed. Rather, use both path-based and railroad-based distance markers and show their locations on user maps for geographic and historic reference.
10. **Community Support.** Many railroad enthusiasts and groups follow and support activities that preserve the history and romance of the railroad. Rail-trail advocates, such as the Rails-to-Trails Conservancy, can also be an important source of citizen support and technical expertise for planning, designing and maintaining rail-trail facilities. Include them as active partners in rail-trail development projects from the outset.
11. **Remoteness.** For a variety of functional reasons, railroad corridors are aligned away from activities, services and cultural experiences that are normally accessible by streets and roads. While this may provide welcome solitude for some rail-trail users, such an experience may be monotonous to others and even threaten personal safety and security. Taking advantage of opportunities to link the corridor back to the roadway system or other bicycle and pedestrian networks as well as providing visual connections to off-trail activity at regular intervals can help overcome these concerns.
12. **Contamination and Deterioration.** Railroad corridors can be the site of environmental contamination due to past spills and freight derailments. The soil along a rail corridor may have become contaminated and an environmental or

Prince Edward Island, Canada



**Retain and use artifacts of prior railroad activity to give rail-trails a "sense of place," strengthening users' understanding of, and connection to, the trail and its surrounding region.**

Swanton – Missisquoi Valley Rail Trail



**Rail-trail advocates can provide an important source of citizen support and technical expertise for rail-trail facilities.**

soils analysis may be required. Similarly, if a former rail yard is to be a part of the future rail-trail corridor, spillage of lubricating oil and other substances may require hazardous waste removal.

Structural deterioration of existing bridge piers, abutments and retaining walls may have occurred. Structures subjected to flowing water or ice flows may be undermined or weakened, requiring a bridge scour analysis to determine their condition and feasibility of reuse.

- 13. Economic Development Potential.** Rail-trail corridors provide recreation, conservation and land value benefits long after the rail line becomes inactive. They are valued for their ability to connect people with places and to enhance the beauty of developed areas. Rail-trails stimulate tourism and recreation-related spending for lodging, food, services, and supplies. They can be the central focus of tourist activities in some communities. Rail-trails preserve critical open space that provides natural buffer zones between development and natural areas. They increase the natural beauty of communities, and have been shown to bolster property values.
- Aside from the utilitarian nature of the rail-trail and its related support facilities, the economic contribution a well-developed and attractive rail-trail can make toward state and local economies should be weighed in planning and design decisions.
- Railroad corridors provide a direct connection to population centers and employment centers. As a result they are well suited to provide direct transportation links independent from the roadway system. In addition, rail-trails can provide long distance off-road connections between points of interest, outdoor recreation activities, and economic development opportunities.

6.1.2 Enabling Legislation

The development of rail-trails has been made possible by legislation at the state (see VSA Title 5, Section 3408) and federal levels (USC, Title 16, Chapter 27, Section 1247(d), also known as “the Trails Act”).

**VSA Title 5, § 3408. Railbanking.** Empowers the Secretary of Transportation to place a state-owned railroad property not economically feasible under present conditions in railbanked status and permits the agency, on behalf of the state, to hold the right-of-way of a railbanked line for reactivation of railroad service or for other public purposes not inconsistent with future reactivation of railroad service.

**USC, Title 16, Chapter 27, Section 1247(d) Interim use of railroad rights-of-way.** The Trails Act directs the U.S. Secretary of Transportation, the Chairman of the Surface Transportation Board, and the Secretary of the Interior to encourage State and local agencies and private interests to establish appropriate trails for interim uses through the provisions of the Railroad Revitalization and Regulatory Reform Act of 1976 in order to preserve established railroad rights-of-way for future reactivation of rail service and to protect rail transportation corridors.

6.1.3 Application

Railroads and rail-trails in Vermont

- In 2000, there were about 1213 km (754 mi) of railroad corridor in Vermont:
- Owned by private companies..... 563 km (350 mi)
  - Owned by the State of Vermont ..... 650 km (404 mi)
  - Leased to railroad operators ..... 526 km (327 mi)
  - Rail-trails ..... 124 km (77 mi)

For a current map of railroad owners, operators and corridor locations refer to [www.aot.state.vt.us/omc/images/rrmap.jpg](http://www.aot.state.vt.us/omc/images/rrmap.jpg). The map includes some, but not all, rail-trail installations.

Although the principal purpose of developing a rail-trail along an inactive or abandoned railway corridor is to provide transportation and recreation opportunities for non-motorized users, rail-trails can also provide significant benefits to the railroad industry. The fundamental precept of the federal railbanking legislation is the preservation of inactive railroad corridor for future railroad use.

Rail-trail development need not be limited to inactive or abandoned corridors. Locating a rail-trail along an active railway (rail-with trail) can safely extend Vermont's network of pedestrian and bicycle facilities and presents the greatest opportunity for future rail-trail development. Vermont is in a unique position because more than 50 percent of Vermont's railroad corridors are currently owned by the State. FHWA's *Rails-with-Trails: Best Practices Report* (2001) suggests that people nationwide are using and enjoying rail-trails located along active rail lines as communities work with railroad operators to develop, operate and maintain rails-with-trails. Existing rail-with-trail installations can be found in Montpelier, Burlington and Newport.

Rail-with-trails provide the added benefit of allowing railroad operators an opportunity to better regulate unauthorized access and therefore reduce trespassing and risk. Rail-trails also improve access for routine maintenance and security patrols as well as for emergency vehicles. Railroad corridors can be used as a utility corridor for fiber optics, gas transmission, water, sewer or other pipelines.

There are four levels of railroad corridor activity and inactivity:

- Active corridors.
- Inactive corridors that remain under the control of the owner and/or lessee or licensed operator (if any).
- Formally abandoned corridors, including those that are in the process of being formally abandoned.
- Railbanked corridor.

**Active corridors.** The principle function of an active rail corridor is the movement of freight and people by rail. Therefore, rail corridor owners and railroad operators may have little interest in welcoming non-railroad activity onto land that is under their control and supervision. Because corridor lessees and owners are usually self-insured, the railroad industry often bears the burden of litigation for incidents and losses that occur on their property.

However, even when a corridor is in active use for railroad purposes, opportunities for rail-with-trail projects may still exist. Base such projects upon needs that are common to the railway operator and rail-trail developer. Mutual needs might include proactive control of chronic trespassing and vandalism, reduction of fatal and serious accidents and crashes, definition of appropriate grade crossing locations, improved access for maintenance and emergency vehicles, and better control of lateral and physical separation from tracks.

**Inactive corridors that remain under the control of the owner and/or lessee or licensed operator (if any).** Under this scenario, active rail use has been discontinued but the corridor remains under the control of the current owner, lessee or licensed operator (if any). Rails and ties may or may not have been removed, but a formal abandonment process has not been initiated with the Federal Surface Trans-

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portation Board. Depending on the prospects for future rail business, the controlling interests may allow the corridor to be developed for rail-trail use.

The most important aspect of a railroad corridor is the land and continuity of the corridor.

**Formally abandoned corridors, including those that are in the process of being formally abandoned.** There exist in Vermont some rail corridors that have been abandoned but not railbanked. The Railroad Revitalization and Regulatory Reform Act of 1976 (4-R Act) allows for transfer of abandoned corridors to public use, including for rail-trails.

The existence of private uses on land, which may or may not be public rights-of-way, complicates planning for rail-trail use on such corridors. In some cases, adjacent landowners may have incorrectly perceived that the railroad had abandoned its property interest in the right-of-way. Or the title to these corridors may have been so unclear so as to result in landowners considering or using the corridor as their own property. Where current rail-trail use is desired along abandoned corridors, legal assistance is often needed to research land titles and acquire permanent easements from landowners.

Property agreements for trolley line corridors, however, have proven to be less secure. Nearly all were constructed along easements, which have long since “extinguished” themselves.

**Railbanked corridors.** The development of rail-trails has been greatly facilitated by the railroad corridor preservation tool known as “railbanking.” Where active rail service has been discontinued, railbanking can help preserve the integrity of a railroad right-of-way for future railroad use by preventing the inactive right-of-way from reverting to the ownership of adjacent landowners.

Railbanking allows inactive railroad corridors to be transferred to qualified managers for interim uses, such as rail-trails, provided that the interim use is consistent with future reactivation of railroad service. Until such time as these rights-of-way are needed for future rail service, rail-trail managers may be required to assume all carrying costs (liability, maintenance and taxes) of the rights-of-way. The interim use may be discontinued at any time to accommodate the revitalization of rail service along any railbanked corridor.

#### 6.1.4 Historic References

A significant difference between a rail-trail and a shared use path is the historical background associated with most rail corridors. There is usually ample information that can be obtained from or with the help of railroad enthusiasts, retired railroad employees, people living near the right-of-way, local museums, and state and local historic preservation organizations, and via the Internet.

If the rail corridor is very old, rail service along the corridor may have been operated by more than one railroad. Knowing the name(s) of the company or companies that provided rail service along the line will facilitate research of the history of the line. It may also suggest a name for the rail-trail. Third, using the right terminology lends credibility to the project and can help garner community support.

A wealth of historical information is usually associated with old railroads. Timetables, photographs, advertising, and logos may be available from railroad historians and collectors. The industry itself produced track charts, bridge drawings, engineering data, valuation surveys, employee’s timetables and maps, all of which (if recovered) can build interest and develop a following of rail-trail supporters.

Frequently, physical structures associated with railroad operation remain after rail service has been discontinued. Such structures can be used to recall aspects of prior railroad activity or used to house services that support rail-trail use. Where railroad depots, sheds or other buildings exist, seek the assistance of a preservationist, architect or structural engineer to assess whether such resources should be preserved. These professionals can help determine how much of the original structure remains and whether potential exists for restoring or remodeling the building. Such buildings are sometimes used as museums. Otherwise, they may be adapted to house a restaurant or café, equipment outfitters, public restrooms or lodging.

Old railway structures or the area in which they are located may be designated or have the potential to be designated as historic by the National Register of Historic Places. It is important to consult the State Historic Preservation Office (SHPO) during the planning phase of the project. Any use of federal funds in a transportation project requires review by SHPO to determine if the project could have an adverse effect on historic resources.

Rolling stock that is no longer in active use can often be acquired to further link the corridor to an earlier era of railroading. Where a boxcar, caboose or locomotive is available, mount such equipment on sections of rail supported by ties and ballast. Contact local rail advisory groups for guidance in restoring railroad cars to their original appearance.

Historic mile markers (distance markers) should not be removed, and if recreated, they should follow the historical railroad numbering system. Where rail-trail-based distance markers are desired along a rail-trail facility, they should be added in addition to the railroad markers. Show both types of markers on a rail-trail map.

## 6.2 Design

The design guidelines for rail-trails are similar to shared use paths but tailored to fit the special circumstances found along railroad corridors. The guidelines include:

- User requirements.
- Widths and clearances.
- Pavement sections and surfaces.
- Cross slope and drainage.
- Grades.
- Side slopes, barriers, bridge railing and fencing.
- Design speed, horizontal alignment and stopping sight distance.
- Roadway crossings.
- Managing motor vehicle access.
- Structures.
- Supplemental facilities.
- Security.
- Lighting (refer to Chapter 9).
- Maintenance (refer to Chapter 10).
- Vegetation and Landscaping (refer to Chapter 9).

### 6.2.1 User Requirements

For user requirements of pedestrians, bicyclists, and in-line skaters, refer to Section 5.3.1. The user characteristics of cross country skiers; snowshoers and snowmobilers are discussed below. The season during which these type users may be

expected on rail-trails where not specifically prohibited generally runs from mid December to early April.

Cross-Country Skiing

Rail-trails are popular with cross-country skiers of all ages and abilities. The activity can range from racing, day-touring, and short excursions, to backwoods bushwhacking. Average ski tourers can travel more than 5 km/h (3 mph) in good weather. An advanced skier can travel at 13 km/h (8 mph) or more and may travel 30 km (20 mi) or longer in an outing.

Design bridges wide enough for tracks and poles, and strong enough to support grooming equipment. Railings should be 1.4 m (4 ft 6 in) high to compensate for snow accumulation.

Snowshoeing

Most casual snowshoers enjoy loop trails that are up to 5 km (3.1 mi) in length over level to moderate terrain. Ungroomed rail-trails are popular among beginners. Snowshoers can share corridors with other rail-trail users but should be to the side where their tracks do not trample ski tracks.

Snowmobiling

Because rail-trails tend to be linear and may extend over great distances, they are attractive to snowmobiling. If snowmobile use is intended or anticipated, consider the guidelines presented in the most recent editions of the Vermont Association of Snow Travelers, Inc. (VAST) *Guide for the Development of Snowmobile Trails* and *Guide for Snowmobile Trail Signing and Placement*.

Well-designed, signed and maintained rail-trails and snowmobile riding areas not only provide enjoyable recreational snowmobiling opportunities but have been proven to significantly reduce the likelihood of a snowmobiler being injured. Statistics indicate that only 10 to 15 percent of snowmobile crashes occur on well maintained and designed rail-trails where as much as 80 to 90 percent of all snowmobile riding takes place.

Vermont Association of Snow Travelers identifies three classes of snowmobile trails. The minimum design parameters for snowmobile trails (including width, vertical clearance and live loads) are based on the type of grooming operations anticipated as shown in Table 6-1.

Table 6-1.  
Snowmobile Trail Classes

| Class | Width         |               | Clear Zone<br>(per side) | Vertical<br>Clearance | Groomer      |                   |
|-------|---------------|---------------|--------------------------|-----------------------|--------------|-------------------|
|       | Preferred     | Min.          |                          |                       | Width        | Weight            |
| 1     | 4.3 m (14 ft) | 3.6 m (12 ft) | 0.6 m (2 ft)             | 3.6 m (12 ft)         | 2.4 m (8 ft) | 3.6 tonne (4 ton) |
| 2     | 3.6 m (12 ft) | 3.0 m (10 ft) | 0.6 m (2 ft)             | 3.0 m (10 ft)         | 2.0 m (6 ft) | 1.8 tonne (2 ton) |
| 3     | 3.0 m (10 ft) | 2.4 m (8 ft)  | 0.6 m (2 ft)             | 2.4 m (8 ft)          | 1.2 m (4 ft) | 0.9 tonne (1 ton) |

Source: *Guide for the Development of Snowmobile Trails* and *Guide for Snowmobile Trail Signing and Placement* (VAST).

Greater trail width may be required on curves than for straight sections. For example, a power groomer unit 2.4 m (8 ft) wide requires a trail tread width of 5.7 m (19 ft) to negotiate a 7.5 m (25 ft) inside turning radius (the recommended minimum). Where the groomer design speed of a trail is higher than 15 km/h (10 mph), or where steeper cross slopes exist, provide additional trail width.



Rail-trails are popular with cross-country skiers of all ages and abilities.



The minimum design parameters for snowmobile trails (including width, vertical clearance and live loads) are based on the type of grooming operations anticipated.

### Managing Winter Rail-Trail Conflicts

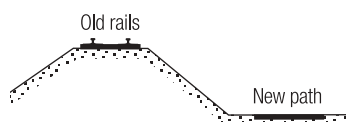
While winter rail-trail users are generally considerate of one another, the speed differential between snowmobiles, skiers and snowshoes can cause conflicts. Where problems are anticipated or later develop, rail-trails lend themselves to being segmented according to user types (i.e., outlying sections of a rail trail could be assigned to snowmobile use, while sections closer to populated areas could be designated for snowshoe and cross country ski use). Further, some sections of the rail-trail might be groomed while other sections remain ungroomed. Speed regulation and enforcement can also help to mitigate conflicts between users. The state established speed limit on state owned rail-trails is 55 km/h (35 mph). Perhaps the single most important safety feature for multi-use of rail-trails is good sight distance.

### 6.2.2 Widths and Clearances

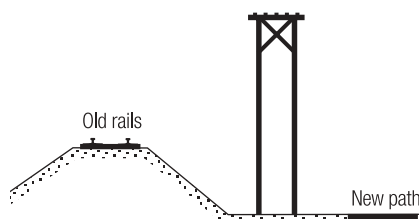
The widths and clearances as specified for shared use paths (refer to Section 5.3.2) also apply for rail-trails and rails-with-trails. Because rail trails are often used by equestrians and snowmobiles, the greater vertical clearances required for those uses often apply. Snowmobile grooming equipment normally requires 3.0 m (10 ft) and equestrian use requires 3.6 m (12 ft). See also VTrans Standard Drawing A-78M, Shared Use Path Typical and VTrans Standard Drawing, Rail-Trail Typical.

#### Strategies for Sub-Optimal Rail-Trail Widths

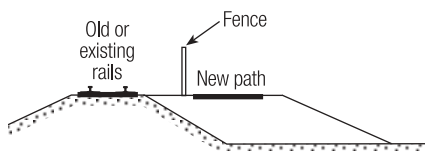
Where adequate widths and clearances cannot be attained, as may be the case where rail corridor berms are extremely narrow or where minimum clearance from an active rail line must be maintained, consider the options illustrated in Figures 6-1 through 6-6:



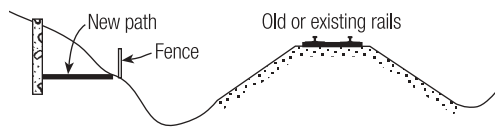
**Figure 6-1.**  
Locate Path at the Bottom of a Slope.



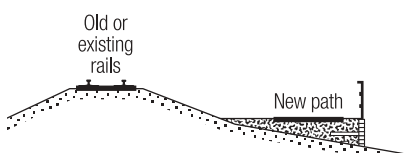
**Figure 6-2.**  
Use an Adjacent Utility Corridor.



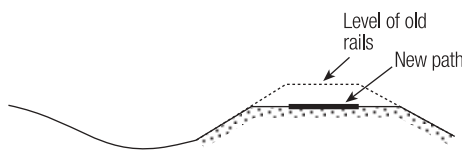
**Figure 6-3.**  
Widen the Berm.



**Figure 6-4.**  
Excavate the Slope and Buttress with a Retaining Wall.



**Figure 6-5.**  
Use a Low Retaining Wall to Increase the Width.



**Figure 6-6.**  
Excavate the Top of the Berm.



Where creating extra width is not an option, it may be possible to proceed with the rail-trail design by restricting uses that require wide tread areas or designing for uses that require less width (i.e., unpaved shared-use path, or narrower ski or snowmobile trails).



**Separation** refers to the space between the edge of improvements associated with a path to the nearest rail face of an adjacent active rail line.

**Setback** refers to the distance from centerline of the nearest active rail line to the edge of path improvements (e.g., signs, fencing or shoulder of a path).

Where creating extra width is not an option, it may be possible to proceed with the rail-trail design by restricting uses that require wide tread areas or designing for uses that require less width (i.e., unpaved shared-use path, or narrower ski or snowmobile trails). Table 6-5 shows the uses that can be accommodated on narrow rail-trails. A more primitive rail-trail that features narrow path widths, natural surfaces, and few or no amenities will also attract fewer rail-trail users than a facility that is highly developed.

Table 6-2.

Designing for Low Levels of Use on Narrow Rail-Trails.

| Available corridor width | Recommended uses*                                    | Shoulder width | Minimum width for bikes and peds | Total         |
|--------------------------|------------------------------------------------------|----------------|----------------------------------|---------------|
| 3.6 m (12 ft)            | Paved shared use path and Class 1 snowmobile trail   | 0.6 m (2 ft)   | 2.4 m (8 ft)                     | 3.6 m (12 ft) |
| 3.0-3.6 m (10-12 ft)     | Unpaved shared use path and Class 2 snowmobile trail | 0.6 m (2 ft)   | 1.8 m (6 ft)                     | 3.0m (10 ft)  |

\* Although a snowmobile trail requires an overall clear width of 4.2 m (12 ft) to 4.8 m (16 ft), depending on the class, trail grooming can result in a build-up of packed snow on both sides of the trail tread area at filled slopes providing a satisfactory usable width in excess of the available corridor width as would be otherwise measured during non-snow conditions. In any case, maintain the overall clear width recommended for each class of snowmobile trail (i.e., minimum clear zone on each side of the trail plus the trail tread area).

Setback and Separation (Rails-with-Trails)

Except as later specified for at-grade railroad crossings in this chapter, separate rail-with-trail improvements (i.e., fences, signs, landscaping, pathways, or other features) from the outside face of the nearest rail of an active railroad track. Where no structures, including fences, occur along the side of a pathway, the edge of the pathway may be as close as 3.3 m (11 ft) from the outside rail where space is at a premium, such as may occur at an undercrossing.

According to FHWA's *Rails-with-Trail: Best Practices Report (2001)* for rail-with-trail corridors, the minimum setback distance should take into consideration the speed and frequency of trains in the corridor, maintenance requirements, separation techniques, existing problem areas and good engineering judgment. However, when adequate setback widths can be achieved along the majority of a corridor and where physical constraints will not allow for the desired setback width, safety should not be compromised. Instead, additional barriers should be considered or additional right-of-way purchased.

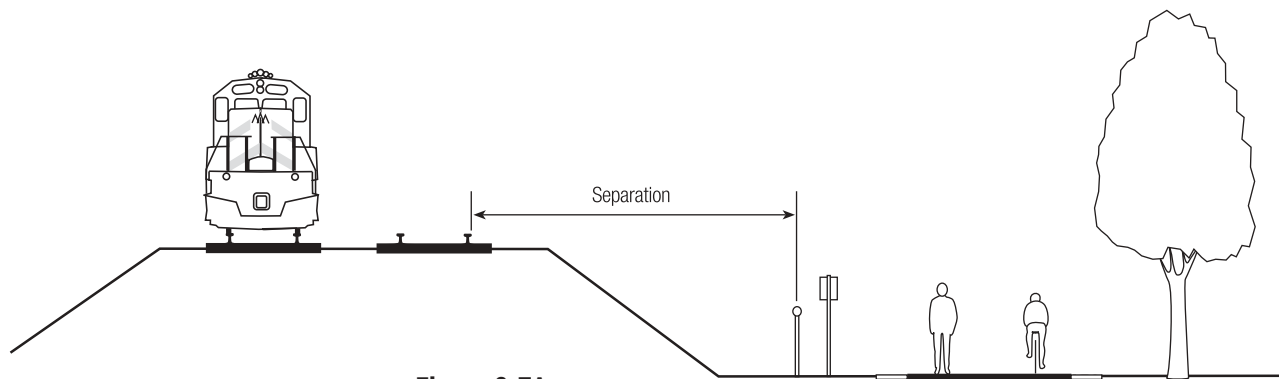
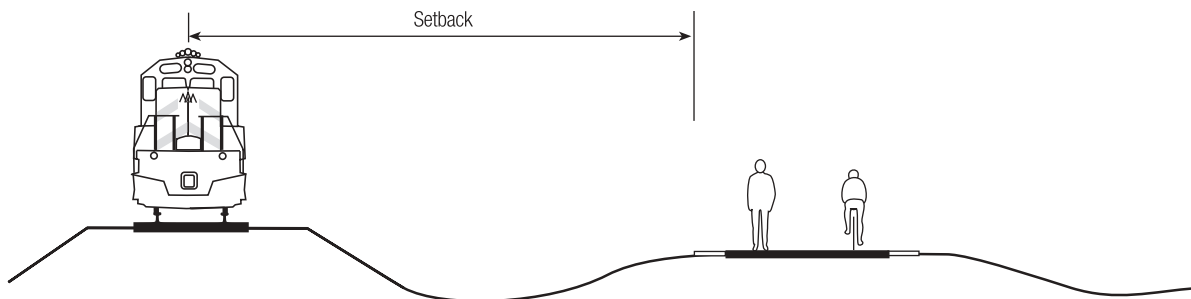
At an absolute minimum the setback must keep rail-trail users outside the “dynamic envelope”, or operating space, of the train. According to the MUTCD, the dynamic envelope is “the clearance required for the train and its cargo to overhang due to any combination of loading, lateral motion or suspension failure”. The dynamic envelope includes the area swept by a turning train.

The recommended minimum rail-with-trail setbacks are shown in Table 6-3 and Figures 6-7A and 6-7B. Exceptions to the recommended setback between a rail-with-trail and track centerlines may be acceptable to the railroad operator or property owner. However, such variations should be negotiated on a case-by-case basis with all affected parties. These possible exceptions are in constrained areas and should only be applied along relatively low speed and volume rail lines.

**Table 6-3.**  
Minimum Separation from Active Rail Line

|                                                                                                        |                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| High density/high speed lines:<br>(11 or more trains per day; maximum<br>speed over 72 km/h (45 m/h))  | Recommended: 7.6 m (25 ft) or more,<br>with fence or other separation technique<br>Minimum: 4.6 m (15 ft), with solid-type barrier |
| In constrained areas, e.g., cut/fill,<br>bridges, trestles                                             | Minimum: 4.6 m (15 ft), with fence or other<br>separation technique                                                                |
| With vertical separation or 3 m (10 ft)                                                                | Minimum: 6 m (20 ft)                                                                                                               |
| Medium density/medium speed lines:<br>(less than 11 trains per day; maximum<br>speed 72 km/h (45 m/h)) | Recommended: 7.6 m (25 ft) or more,<br>Minimum: 4.6 m (15 ft), with adequate separation<br>technique                               |
| In constrained areas, e.g. cut/fill,<br>bridges, trestles                                              | Minimum: 3.3 m (11 ft), with fence or other separation<br>technique                                                                |
| Extensive history of trespassing<br>(>100 persons per day)                                             | Minimum: 3.3 m (11 ft), with fence or other separation<br>technique                                                                |
| Low density/low speed branchlines:<br>(less than one train per day; maximum speed<br>56 k/h (35 m/h))  | Recommended: 7.5 m (25 ft) or more<br>Minimum: 3.3 m (11 ft) (RWT to serve as maintenance<br>access)                               |
| In constrained areas, e.g., cut/fill,<br>bridges, trestles                                             | Minimum: 3.3 m (11 ft), with fence or other separation<br>technique                                                                |

Source: *Rail-with-Trails: Lessons Learned – Draft*; Table 5.2 Minimum Setbacks  
with modifications by VTrans Rail Division.

**Figure 6-7A.**  
Separation, Rails-With-Trails.**Figure 6-7B.**  
Setback, Rails-With-Trails.

*Rail-Trails and Rails-With Trails***6.2.3 Pavement Sections and Surfaces**

Except as discussed below, the pavement sections and surfaces as specified for shared use paths (refer to Section 5.3.3) also apply for rail-trails. As with shared use paths, the surface of a rail trail must comply with ADAAG requirements that it be firm, stable and slip-resistant. VTrans has developed a specification (as shown in Table 6-4) for unpaved surfaces that meet this requirement. See also VTrans Standard Drawing, Rail-Trail Typical.

| Sieve Designation (Metric) | Sieve Designation (English) | Percentage by mass (weight) passing square mesh sieves |
|----------------------------|-----------------------------|--------------------------------------------------------|
| 9.5 mm                     | 3/8 inch                    | 80-100                                                 |
| 4.75 mm                    | #4                          | 60-90                                                  |
| 1.18 mm                    | #16                         | 20-40                                                  |
| 600 µ                      | #30                         | 14-30                                                  |
| 300 µ                      | #50                         | 10-25                                                  |
| 150 µ                      | #100                        | 8-15                                                   |
| 75 µ                       | #200                        | 0-8                                                    |

Ohiopyle, PA



**If track and ties have been removed, a sub base of ballast, and in some cases a sub-ballast layer of cinders, may still remain above a sub grade of natural soil or compacted fill.**

**Railroad Beds**

Railroads beds were constructed to support loads as heavy as 13.6 ton (30,000 lb) per train wheel traveling at 50 mph. Obviously, design loads for rail-trail use will be significantly lighter — on the order of 10,000 lb per gross vehicle weight traveling at 15 mph for heavy maintenance and emergency vehicles.

**When the Tracks and Ties have been Removed**

If track and ties have been removed, a sub base of ballast, and in some cases a sub-ballast layer of cinders, may still remain above a sub grade of natural soil or compacted fill.

A common question relating to ballast is should it be kept in place, or should some or all of it be removed? The answer depends on a number of factors, which include:

- Has the corridor been railbanked for possible return to future railroad use?
- Will toxic substances be uncovered if the ballast is disturbed?
- Have crossties left their imprint (“track-tie memory”) in the ballast?
- Do drainage problems exist at the sub base or sub grade levels?

To answer these and other questions, the expertise of a structural or geotechnical engineer may be required.

Where possible, it is generally a good idea to leave as much as the ballast in place to disperse the weight of the rail-trail’s wearing surface and to promote drainage. A minimum of 200 mm (8 in) of ballast used as sub-base is recommended. If track-tie memory is an issue, remove the top layer of ballast and regrade, recompact and reconstruct the remaining ballast to an elevation that can support the rail-trail’s design load. Retain excess clean ballast, preferably at the construction site for reuse as needed for either construction or maintenance.

**When the Tracks and Crossties Remain**

The preferred technique that has been used in Vermont is to bring in fill sufficient enough to cover over the ties and rails. See Figure 6-8. This technique should be viewed as a short-term stopgap measure only. The major disadvantage of this approach is that the rail-trail is more prone to washouts and exposure of the under-

lying rails and ties can create a hazard for rail-trail users and will require additional maintenance. However, such a compromise may be necessary to obtain local and political support for development of a rail-trail. It should be noted that any remedy that calls for filling in and around the ties and rails will accelerate the deterioration of the ties and rails. The maintenance costs and benefits should be fully weighted before selecting either of these options.



**Figure 6-8.**

Creating a Temporary Pedestrian Path when Crossties and Rails Remain

### **Geotextiles**

Consider the use of geotextiles (fabric mats that increase the stability of subsurface materials) in rail trail construction. For aggregate surface rail trails, place the geotextile between the sub base and the surface course. For paved rail trails, place the geotextile between the sub grade and the sub base if this is a practical alternative. Geotextile can help control vegetation on unpaved rail-trails and prevents the surface material from migrating down into the ballast and prevents the ballast from migrating up to the surface of the rail-trail.

### **6.2.4 Grades**

Because the grades of a railroad corridor rarely exceed 2.5 percent (1:40), the grade requirements of ADA and as specified for shared use paths should be easily met. However, grades may be an issue when providing access from locations adjacent to rail-trail facilities. In such cases, follow the recommendations for shared use paths (refer to Section 5.3.4).

### **6.2.5 Cross Slope and Drainage**

The recommendation for cross slope and drainage as specified for shared use paths (refer to Section 5.3.5) also apply for rail-trails. See VTrans Standard Drawing, Rail-Trail Typical.

### **6.2.6 Side Slopes, Bridge Railings, Barriers and Fencing**

One main difference between shared use paths and rail trails is that the supporting structure of fill and ballast usually exists on a rail bed prior to its development as a trail. It is often impractical and costly to add material to existing railroad bed fill slopes. Additionally, many rail lines traverse rural, vegetated areas with native vegetation growing along the sides of the rail bed. This results in trails that meet minimum path widths, but often lack preferred shoulder and lateral clearance widths. Steeper than 1:3 side slopes are often present, however many rail trail are unpaved, resulting in slower user speeds. Existing vegetation and the presence of recovery areas between the edge of the path and the top of slope may eliminate the need for barriers in many cases.

The side slopes, bridge railings, barriers and fencing as specified for shared use paths (refer to Section 5.3.6) also apply for rail-trails. Pay particular attention to guidelines for recovery areas for unpaved paths as shown in Table 5-2. See also VTrans Standard Drawing, Rail-Trail Typical.

### **Sub-Optimal Side Slope**

Where steep side slopes are encountered and the minimum requirements for recovery areas described for shared use paths cannot be attained (refer to Section

***Rail-Trails and Rails-With Trails***

5.3.2 and 5.3.6), consider the following options:

- Use more forgiving materials (e.g., vegetation and grass) along steep side slopes.
- Provide landscaping along the slopes of the berm to limit the extent of accidental falls.
- Eliminate the need for a railing or barrier (e.g., move the path to bottom of slope or widen berm).
- Install warning signs and pavement markings.
- Use a combination of these techniques.

**Special Considerations for Barriers on Rails-with-Trails**

The primary objective of a barrier between a rail-trail and active rail line is to separate and channelize users to existing or new formal grade or grade-separated crossings. Occasional breaks in such a barrier are provided for environmental requirements, safety and access. Depending on the level of separation required, barriers typically consist of fencing, vegetation, grade or ditches.

Because the development of a rail-with-trail may impact railroad operation, the barrier type, material and location should be determined jointly among the interested parties. According to *FHWA Rails-with-Trail: Lessons Learned (2002)* over 70 percent of existing rail-with-trails in the United States are separated from tracks by some kind of barrier, including fencing (34 percent), vegetation (21 percent), grade (16 percent), or drainage ditch (12 percent). The fencing style varies from chain link, woven wire mesh, wrought iron, vinyl, and steel picket to wooden rail.

**Vegetation, grades and ditches.** Vegetation, grades and ditches may be used as a barrier between the rail-trail and an active track when:

- The rail-trail (not including shoulder) is located farther than 7.6m (25 ft) from the nearest rail face, or
- The vertical separation between the surface of the rail-trail and the track is greater than 3 m (10 ft).

**6.2.7 Design Speed, Horizontal Alignment and Stopping Sight Distance**

The design speeds, horizontal alignment and stopping sight distance specified for shared use paths (refer to Section 5.3.7) also apply for rail-trails. These requirements should be easy to meet for rail-trails that occupy rail corridors.

**6.2.8 Crossings****Roadway Crossings**

The recommended treatments for roadway crossings as described for shared use paths (refer to Section 5.3.8) also apply to rail-trails.

When an at-grade rail-trail crossing intersects a roadway less than 105 m (350 ft) from a highway intersection, consideration should be given to diverting the crossing to the highway intersection and treat it as a parallel path crossing or pedestrian crossing. However, this may not be possible if the area between the crossing and intersection is heavily developed, as in a village or urban area.

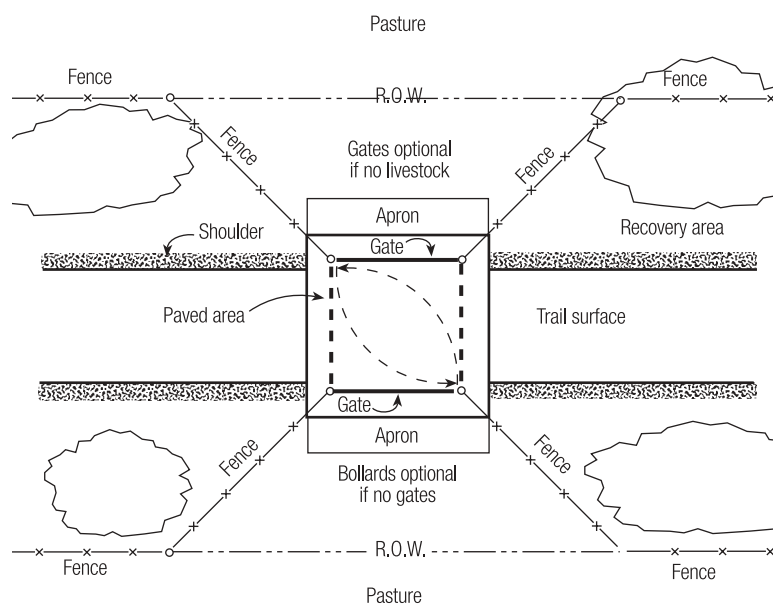
**Agricultural Crossings and Private Access**

A thorough analysis of land records and valuation sheets will likely reveal where adjacent landowners have a legal right to cross a railroad corridor, even after railroad service has been discontinued. The principle design issues are:

- Ensuring that access for the landowner is perpetuated throughout the year (including winter).

- Safety of rail-trail users.
- Assuring convenience for adjacent landowners.
- Discouraging trespassers.
- Assigning responsibility for rail-trail maintenance.

**Agricultural crossings.** Farm crossings usually involve the movement of livestock and equipment from one side of the rail-trail corridor to the other. Therefore, it may be necessary to provide a way to guide livestock across the rail-trail while preventing escape along the rail-trail. A simple fence and gate arrangement (as shown in Figure 6-9) can be provided to accomplish both objectives as long as the gates are capable of operating without difficulty even during winter conditions. Pave the rail-trail crossing with asphalt for the full width of the crossing. Do not use cattle guards on or near any part of the rail-trail surface. Also, special attention should be given to keeping the rail-trail clear of debris and manure.



**Figure 6-9.**  
Agricultural crossing.

**Private access.** The only way to access some camps, vacation homes, full-time residences and businesses is by crossing a rail-trail corridor. For the property owner, trespassing by rail-trail users may be a concern. For the rail-trail user, damage to the travel surface or accumulation of excess gravel can be an issue. “No trespassing signs” and notices in rail-trail literature can help remind rail-trail users not to trespass.

The responsibility for providing and installing “no trespassing” signs should be a part of an access agreement between the rail-trail management entity and the individual landowner. To maintain uniformity in signs, the rail-trail manager may be willing to accept the responsibility for providing and installing the signs in exchange for the property owner keeping the rail-trail clear of excess gravel and stones. Where cross traffic creates a chronic maintenance problem, consider paving the crossing with asphalt.

### 6.2.9 Managing Motor Vehicle Access

The recommended treatments for managing unauthorized motor vehicle access as described for shared use paths (refer to Section 5.3.9) also apply to rail-trails.

*Rail-Trails and Rails-With Trails***6.2.10 Structures****Bridges**

The recommended treatments for bridges, railings and approaches as described for shared use paths (refer to Section 5.3.10) also apply to rail-trails. However, consider the following special conditions.

**Existing bridges.** A structural engineer should evaluate existing railroad bridges for structural integrity to ensure they are capable of carrying the appropriate design loads. The standard practice for path bridges is to design for H-10 loading.

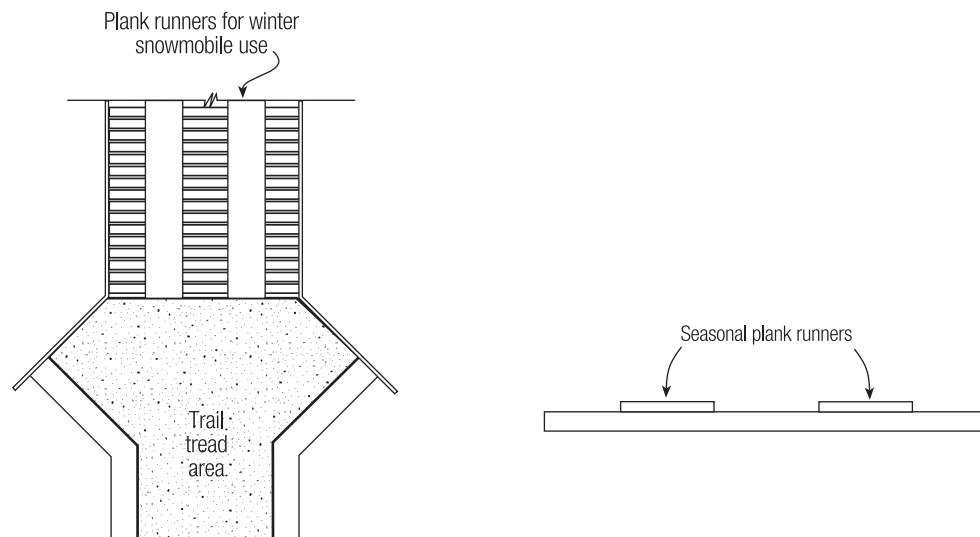
If bridge abutments or piers have been subjected to flowing water during their lifetime, a bridge scour analysis may also be necessary.

An historic preservation specialist should assess whether the bridge has historic or architectural significance.

If the bridge lacks decking suitable for rail-trail use, new decking capable of supporting the design load must be installed. Where decking is used, it should be laid transversely across the bridge (90 degrees to the direction of bicycle travel) to avoid joints that may trap a bicycle wheel.

Provide gaps of 3-6 mm (0.125-0.25 in.) between decking members to allow for drainage. Where snowmobile use is expected or permitted, temporary seasonal plank runners may be laid over the permanent deck to protect it from the metal cleats commonly found on snowmobile tracks. Where used, the runners should be laid parallel to the path of travel of the snowmobiles as shown in Figure 6-10.

Because runners can interfere with cross-country skiers, the management plan for operation of the rail-trail should include a plan to resolve user conflicts early on. Plank runners should be promptly removed in the spring.



**Figure 6-10.**  
Seasonal Plank Runners.

**Adapting historic bridges for reuse.** VTrans operates an Adaptive Use Bridge Program as a part of their Historic Bridge Program. The purpose of the Adaptive Use Program is to preserve, restore, relocate and reuse historic bridges. The success of the program depends upon a viable system for preserving bridges that can no

longer serve highway uses at their existing locations. To meet this need, qualifying bridges are adapted to alternative transportation uses at their existing sites or will be relocated for alternative transportation uses such as pedestrian and bicycle paths, snowmobile trails, recreational sites, or historic sites.

Where reuse of an historic bridge appears feasible, consult the VTrans Adaptive Use Bridge Program manual for information about (a) methods for identifying and selecting new sites for historic bridges; (b) preparation of rehabilitation plans; (c) development of contracts for different aspects of each project; (d) cleaning and painting bridges; (e) names of contractors qualified to perform work; (f) coordination regarding materials and labor supplied by the Vermont Department of Corrections; (g) archival documentation; and other appropriate details.

**New bridges.** If old bridges have been removed, it may be possible to utilize existing abutments and piers (if any) to support a new structure. Again, a structural engineer should evaluate the structural integrity of any remaining supports and a scour analysis should be performed (where applicable) to ensure that the piers and abutments are capable of carrying the appropriate design loads.

Where no previous supports remain, a new bridge will have to be provided. The first choice should be to determine whether an adaptive use bridge, no longer suitable to carry modern highway traffic, is available, possibly from the inventory of previously stockpiled bridges. Designed to carry heavier loads than may be required for the rail-trail (notwithstanding use by maintenance and emergency vehicles), adaptive use bridges may cost less to refurbish than it would cost to erect a new bridge. Also, adaptive use bridges are likely to be more in keeping with the historical context of the railroad corridor.

If a new bridge must be considered, the type of superstructure to be used will depend on specific site conditions including: whether a waterway, roadway or railway will be bridged; the length of span required; and access to the construction site. Refer to Section 5.3.10, Shared Use Paths for a complete discussion of design consideration for structures.

As a general rule, the following types of structures, based on spans, have proven successful in other areas.

- 0 to 4.5 m (0 to 15 ft): Consider single span, treated timber structures, using standard dimension lumber, with timber crib abutments.
- 4.5 to 10.7 m (15 to 35 ft): Consider multiple span, treated timber structures, using standard dimension lumber or laminated timber superstructure components with timber crib or timber pile bent substructures.
- 10.7 to 27.4 m (35 to 90 ft): Consider prefabricated metal superstructures, or steel or prestressed concrete girders with timber or concrete decks, reinforced concrete on spread footings or piles.
- 27.4 m (90 ft) or more: Consider prefabricated metal superstructures and reinforced concrete on spread footings or piles.

**Overpasses and underpasses.** The recommended treatments for overpasses and underpasses as described for shared use paths (refer to Section 5.3.10) also apply to rail-trails. Where a rail corridor has been railbanked, preserve a vertical clearance as required for railroad operation: 7 m (23 ft) above the top of the highest rail.

Where an existing railroad bridge has been removed, another approach is to use a corrugated metal culvert and backfilled material to support the rail-trail while providing access under the rail-trail.

*Rail-Trails and Rails-With Trails***6.2.11 Supplemental Facilities and Amenities**

To enable a rail-trail to develop to its full potential, consider a broad range of support facilities (refer to Section 5.3.11). The extent to which these facilities are made available depends on:

- Rail-trail setting and uses.
- Intensity of use.
- Maintenance.
- Utilities and infrastructure required.

Refer to Section 9, Landscaping and Amenities, for a discussion of supplemental facilities and other amenities.

Ohioople, PA



**Rail-trail amenities — like this bird feeder — provide opportunities to acquaint users with flora and fauna of the area.**

Confluence, PA



**Rail-trails can provide access to natural areas for activities unrelated to walking or bicycling. This stairway allows anglers to reach a popular fishing hole.**

**6.3 Additional Resources**

Users of this manual are encouraged to consult the following resources for the broadest coverage of issues relating to the planning and design of rail-trails and rails-with-trails.

- *Guide for the Development of Bicycle Facilities*, American Association of State Highway and Transportation Officials (AASHTO), Washington, DC. 1999.
- *Guide for the Development of Snowmobile Trails*, Vermont Association of Snow Travelers, Inc. 2000-2001.
- *Guide for Snowmobile Trail Signing and Placement*, Vermont Association of Snow Travelers, Inc. 2000-2001.
- *Designing Sidewalks and Trails for Access: Best Practices Design Guide*, Federal Highway Administration (FHWA), 2001.
- *Manual for Railway Engineering*, American Railway Engineering and Maintenance-of-Way Association, 8201 Corporate Drive, Suite 1125, Landover, MD 20785, Phone: 301-459-3200, Fax: 301-459-8077, [www.arena.org](http://www.arena.org). 2001.
- *Rail-trails and Liability*, Rails-to-Trails Conservancy, 1100 Seventeenth Street, NW, Washington, DC 20036, Phone: 202-331-9696, Fax: 202-331-9680, [www.railtrails.org](http://www.railtrails.org). 2000.
- *Rails-with-Trails*, Rails-to-Trails Conservancy, 1100 Seventeenth Street, NW, Washington, DC 20036, Phone: 202-331-9696, Fax: 202-331-9680, [www.railtrails.org](http://www.railtrails.org). 2000.
- *Rails-with-Trails: Best Practices Report*, Federal Highway Administration (FHWA), 2001.
- *Rails with Trails: Design, Management and Operation Characteristics of 61 Trails along Active Railroads*, November 2000, Rails-to-Trails Conservancy.
- *Trails for the Twenty-First Century*, Rails-to-Trails Conservancy, 10<sup>th</sup> floor, 1100 Seventeenth Street, NW, Washington, DC 20036, Phone: 202-331-9696, Fax: 202-331-9680, [www.railtrails.org](http://www.railtrails.org). 2001.